

Ziye Lin

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<https://github.com/zylin1051>

EDUCATION

University of Waterloo

Master of Mathematics, Biostatistics

Waterloo, ON, Canada

Aug 2025

- Supervisor: Changbao Wu
- GPA: 86/100
- Research paper: Imputation-based Approaches to Causal Inference
- Award: 2024 W.F. Forbes Entrance Award

University of Waterloo

Bachelor of Mathematics, Statistics

Waterloo, ON, Canada

Aug 2023

- GPA: 89/100
- Award: 2020 President's Scholarship of Distinction

TEACHING DOSSIER

Teaching dossier available [online](#).

TEACHING EXPERIENCE

Tutorial Leader – [STAT 230 \(Probability\)](#)

Department of Statistics and Actuarial Science, University of Waterloo

Waterloo, ON, Canada

May 2025 – Aug 2025

- Delivered tutorial sessions regularly to approximately 40-50 students
- Clarified concepts and reviewed practice problems
- Interacted with students to answer questions and provide support

Guest Lecturer – [STAT 231 \(Statistics\)](#)

Department of Statistics and Actuarial Science, University of Waterloo

Waterloo, ON, Canada

Mar 21, 2025

- Delivered a guest lecture to a class of 28 students
- Covered Gaussian response models and model diagnostics
- View the [lecture notes](#)

Grader – Statistics Courses

Department of Statistics and Actuarial Science, University of Waterloo

Waterloo, ON, Canada

Sep 2024 – Aug 2025

- Provided written feedback to support students' learning
- Graded assignments and exams:
 - STAT 230 (Probability) (Sep 2024 – Dec 2024 and May 2025 – Aug 2025)
 - [STAT 330 \(Mathematical Statistics\)](#) (Sep 2024 – Apr 2025)
 - [STAT 331 \(Applied Linear Models\)](#) (May 2025 – Aug 2025)
 - [STAT 332 \(Sampling and Experimental Design\)](#) (Jan 2025 – Apr 2025)

Math Tutor – [MATH 237 \(Calculus III for Honours Math.\)](#)

Mathematics Tutoring Centre, University of Waterloo

Waterloo, ON, Canada

Jan 2023 – Apr 2023

- Conducted one-on-one in-person tutoring sessions with students
- Clarified course material and provided strategic hints for assignments

TEACHING AND PROFESSIONAL DEVELOPMENT

University Mathematics Teaching Techniques ([MATH 900](#))

Faculty of Mathematics, University of Waterloo

Waterloo, ON, Canada

Jan 2025 – Apr 2025

- Completed formal training in university-level mathematics instruction
- Studied course design, lecture effectiveness, assessment strategies, and student engagement
- Delivered three peer-reviewed teaching sessions on probability theory and statistical modeling
- Delivered a guest lecture to a live undergraduate class: STAT 231 (Statistics)
- Designed an assignment question with marking rubric
- Authored a formal teaching statement
- By completion, qualified to be appointed as a graduate instructor in the Department of Statistics and Actuarial Science for undergraduate courses

Teaching Workshops

Centre for Teaching Excellence, University of Waterloo

Waterloo, ON, Canada

Jan 2025 – Jun 2025

- Participated in four professional development workshops for post-secondary teaching
- [CTE 1106](#): Explored effective feedback strategies for teaching, focusing on phrasing, timing, utility, and emotional impact. Developed practical approaches to enhance student learning and engagement through constructive feedback
- [CTE 1111](#): Focused on designing effective tutorials in STEM disciplines. Covered strategies for planning, running, and assessing tutorials with an emphasis on active learning and student engagement. Included hands-on development of tutorial plans and peer feedback
- [CTE 1217](#): Explored the principles and benefits of active learning. Gained practical strategies and a toolkit of activities to increase student engagement, compare active learning with traditional lecturing, and design and implement interactive classroom activities
- [CTE 1234](#): Examined common causes of low teaching confidence among graduate instructors and explored mindset shifts and practical strategies to strengthen confidence and effectiveness in the classroom

SELECTED PROJECTS

Imputation-based Approaches to Causal Inference

Master's Research Paper, University of Waterloo

Waterloo, ON, Canada

May 2025 – Aug 2025

Supervisor: Changbao Wu

- Explored imputation approaches in causal inference
- Reviewed Propensity Score Matching, Inverse Probability Weighted Estimators, AIPWE
- Discussed imputation methods for missing data and missing data mechanisms
- Investigated the performance of regression and nearest neighbor imputation (in R) via simulations to assess bias and efficiency under different settings

Bayesian Mortality Modelling — Seminar and Review

Actuarial Science Topic Project, University of Waterloo

Waterloo, ON, Canada

Jul 2025 – Aug 2025

Topic: Managing Longevity Risk

- Paper: [Bayesian mortality modelling with pandemics: a vanishing jump approach](#)
- Delivered a graduate-level seminar on the paper's central ideas, research methods, and findings
- Conducted a peer-review process and authored a formal review report
- [Seminar slides](#) and [review report](#)

- Modelled the association between the observed radial velocity and the position in the sky from which it is observed for NGC 7531
- Applied ordinary least squares, regularized least squares with ℓ^2 -norm penalty, cubic polynomials, and cubic B-splines in R
- Assessed model performances by AIC, training error, and estimated mean prediction squared error via cross-validation
- Source Dataset: [The Elements of Statistical Learning](#) → Data → Galaxy
- [Project Report](#) and [Appendix \(Code\)](#)

ADDITIONAL INFORMATION

- Programming: R, Python, SQL, SAS
- Typesetting: $\text{\LaTeX}$, Markdown
- Language: English (fluent); Mandarin (native); Cantonese (native)
- Work Eligibility: Eligible to work full-time in Canada; applied for PGWP.